

Survey of Simulation Capabilities of the IBM WebSphere Business Modeler Business Process Modeling Tool on the Example of Processing a Loan Application

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Abstract. *Business process modeling tools that support simulation are a powerful means of improving business processes. With a view to avoid the risk of failure and unnecessary costs, current state process models are developed as an alternative to conducting real-world business process modification. Further attempts are made at improving those models. The aim of this paper is to show some of the model simulation and analysis capabilities of the IBM WebSphere Business Modeler. IBM is one of the leading tool vendors in this area and its market share is constantly rising. Case analysis is made on the example of Processing a loan application in one of largest Croatian banks. There are recognized and systematized simulation capabilities of the tool, which supports the Business Process Modeling Notation (BPMN).*

Keywords. Business Process Modeling, IBM WebSphere, Business Process Redesign and Improvement, Simulation, BPMN.

1. Introduction

In the current economic conditions marked by the global recession, business organizations need to adapt to new circumstances in the market, as well as respond to major competition pressures. One of the ways in which this can be achieved is through business process improvement and/or reengineering. Issues related to business process improvement and/or reengineering which tend to be the most common causes of failure are: lack of creativity in modeling, the cost of implementation of new processes, inability to recognize their dynamic nature, and, most importantly, to accurately predict the results arising from the implementation of certain radical changes. Errors which may occur upon process redesign can be recognized only after the redesigned processes have been implemented which eventually makes it impossible to correct

wrong decisions or at least makes it extremely costly to do so. Business process modeling is the key technique enabling successful business process reengineering.

Modeling substitutes live experimenting, which is expensive and tends to be time-consuming, requires costly repetition of experiments, and can incur hazards [1]. Developed model initially displays the current state 'AS IS' process, after which the model is improved by designing a future process of the so-called 'TO BE' model. In doing so, both simulation modeling itself and the tools for business process modeling with simulation capabilities play a significant role. Simulation modeling is techniques for constructing and implementing a real-system model with a view to examine its behaviour without distractions from the environment [2].

There are several features that make simulation suitable for business process modeling: a) designed simulation models can be easily modified if changes in the real system should occur and can therefore be used as a decision support tool; b) simulation models can comprise the behaviour of both human and technical resources in a given system; c) simulation models can incorporate the stochastic nature of business processes as well as the accidental behaviour of their; d) the visual interactive features of many simulation packages available on the market enable a graphical display of dynamic behaviour of model entities and show dynamic changes within formulated processes [3].

As these requirements present a significant challenge to modeling, specific modeling tools have been devised as an adequate means for meeting the simulation modeling requirements as well as those imposed by business process modeling. Goal of this paper is to systematically present capabilities of IBM WebSphere which possibilities were examined in detail.

2. Trends in business process modeling

Over the last few years business process modeling as well as business process management have been a very popular theme in scientific and business practice. Lots of modeling techniques like UML, IDS Sheer ARIS EPC, BPMN and others are used and constantly developed to enable business analysts who improve business processes and IT experts who deploy them to deliver a better value.

Since its 2004 version 1.0 until today Business Process Modeling Notation (BPMN) has become a standard for all the major process modeling software tools vendors like IDS Sheer, IBM, Oracle, Microsoft, SAP AG and others [4]. No other notation has seen such an uptake in such a short time as BPMN has. It is widely supported by both free and commercial process modeling tools integrated into the curriculum of education providers [5]. Developed by the Object Management Group, (OMG), the reputable international standards organization that supports all major standards in IT industry (UML, CMM, etc.), BPMN is intended to serve as notation that is understandable to all business users [6]. According to a leading research in business process modeling trends survey conducted by the BPTrends Association in February 2010 for the year 2009, BPMN is the leading standard that organizations are interested to adopt (22% organizations in 2005, 41% in 2007, 51% in 2009). The sample included 595 business process experts worldwide [7]. From its acceptance the BPMN notation implies the usage of the standard Business Process Modeling Language (BPML) and can be used to transform a static business process model into Business Process Execution Language (BPEL). In this way a process model can become executable and easier to implement with the support of information technology in practice. However, the notation is being constantly developed, so the official BPMN Version 2.0 is expected in 2010.

The following section contains an overview of some relevant business process modeling tools with simulation capabilities and integrated support for BPMN standard.

3. Contemporary tools for business process modeling and simulation

A growing interest in business process modeling over the last decade has given rise to a wide offer of business process modeling tools on

the software market. Such tools can be used to predict characteristics that cannot be directly measured and can also predict economic and performance data that would otherwise be too expensive or impossible to acquire [8].

Owing to fierce competition, potential customers can choose from an ample supply of tools, wherein the issue of selecting the most appropriate software is raised [9]. According to the latest survey from February 2010, leaders in Gartner 2010 Magic Quadrant are currently IDS Scheer (ARIS), IBM (WebSphere Business Modeler), Metastorm (Metastorm BPM), Mega (Mega Modeling Suite), iGrafx, Microsoft (Visio) [10]. These are all proven vendors with high degrees of functionality and market penetration. Their market share in 2009 was: Microsoft Visio 36%, ARIS 18%, IBM 17%, Metastorm 3.4%, Mega 3% and iGrafx 3.5%. The market share does not entirely represent the power and suitability of these tools but it shows current trends.

Table 1 shows the data on revenue share of the Business Process Analysis tool market over the past few years [10], [11], [12]. The low cost and ease of use of Microsoft's BPA tool, Visio, have resulted in Microsoft having by far the largest number of actual implementations of any process modeling tool.

Table 1. Revenue share of the Business Process Analysis tools market.

Business Process Analysis tool	Revenue share of the Business Process Analysis tools market		
	2007	2008	2009
Microsoft	30%	34%	36%
IDS Scheer	15%	15%	18%
IBM	9%	16,4%	17%
iGrafx	3%	3.5%	3.5%
Mega	4%	2.6%	3%
Megastorm	-	3.4%	3.4%
Telelogic	7%	-	-
Proforma	3%	-	-
Casewise	3%	3,4%	-
EMC	3%	-	-
Others	23%	22%	19%

Data from these studies show that in the past few years, IBM WebSphere Business Modeler tool has been one of the leading tools in this field and that its market share is constantly growing.

4. IBM WebSphere Business Modeler

IBM WebSphere Business Modeler is an integrated tool for business process modeling,

analysis and implementation, oriented towards improving the overall business process management. Its primary purpose is to assist experts in different fields, to collaborate in clearly defining business models and assessing their efficiency. The tool makes it possible to develop models of new or the existing processes, implement business processes, track the key indicators of process performance, undertake corrective action and initiate changes aimed at continuous improvement of business processes [13]. With WebSphere Business Modeler, a business analyst can fully visualize, understand, document, test, and share business processes. Process simulations can be performed to identify bottlenecks and inefficiencies, and define key performance indicators and business metrics for use in WebSphere Business Monitor [14]. For using this tool the standardized BPMN is used.

WebSphere Business Modeler is the key tool in the IBM BPM Suite. Owing to its process modeling capabilities, the tool makes it possible to create a hierarchical process structure as well as simulate and improve each process separately. In addition, it enables the simulation and restructuring of the set of integrated business processes as a business technology, i.e. business

process reengineering. Business processes can be analyzed from several aspects: process duration, activities waiting time, resource operation time spent, total cost of process execution, total revenue and profit to be yielded in the process.

5. Research and methodology

In order to examine the application and capabilities of the tool described above an example of the process entitled Processing a loan application was developed and adapted. The example was modelled on the basis of a previously examined business process and real world data collected in one of major Croatian banks. The purpose of this case study example is to provide insight into the real process on a simplified process model. The model is run with the data collected from different positions and with the available statistical data obtained from the bank. The current state process model (AS IS) in Fig. 1 shows five activities or worksteps in the loan-processing process: accepting the loan application, considering the loan application, approving or denying the loan application and loan disbursement.

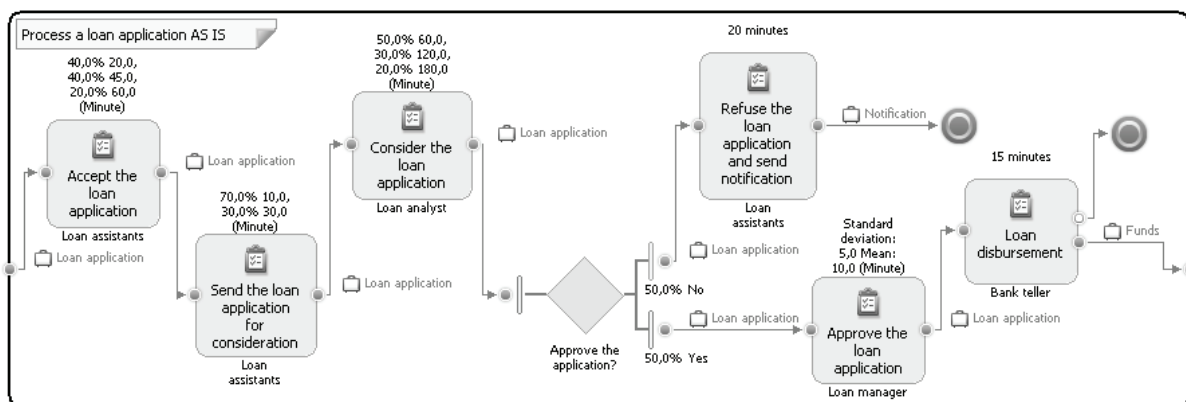


Figure 1. Business process – Processing a loan application AS IS

By investigating the gathered real-world process data required for the operationalization of the model, the average activity duration time was recorded. For individual activities the average duration time, displayed above each activity in the model, see Fig. 1, was defined stochastically and/or deterministically. Time is different for every type of loan (residential, cash, car purchase) and amount of requested loan. During the simulation of the model the average loan application arrival time of 1 hour and 30 minutes was used, with the bank opening times 7:00am-3:00pm (8-hour opening times, closed on

weekends). The activities in the model were assigned the resource operation time, whose gross labour price per hour. Gross labour price for role were: Bank teller 35 kn/h, Loan assistants 55 kn/h, Loan analyst 65 kn/h, and Loan Manager 85 kn/h. Prices were extracted from average labour price in Croatia for year 2009 and increased in percentage in accordance with structured positions and educational level. Data are just example and they do not present real bank data.

Based on the analysis and tracking of the average number of applications monthly

accepted by the branch offices, it was calculated that a single branch office processes around 129 loan applications per month. This amounts to 1548 applications per year (the figure refers to applications accepted within one shift by a single application assistant). It amounts to approximately 6 applications per day. To verify the correctness of the model simulations were performed over 12 months. For each approved loan application there is a 2% loan processing fee charged by the bank. In the present example, for an average loan of approximately 10.000 euros (73.000 kn) the fee amounts to 1.460 kn. Upon performing the simulation on the AS IS model the simulation results were compared against the actual data. It was established that the simulated model corresponds with the real-world process, with the deviation from the real-world process approximately amounting to a mere 0.54%. The

results obtained in the performed simulations of the AS IS model are shown in Table 2.

Table 2. Simulation results in the AS IS state

Process	Duration	Cost	Revenue	Profit
AS IS	2 hrs 59 min	156,62 kn	730,00 kn	573,38 kn

Based on the verified AS IS process model the improved TO BE model was developed by changing the possibilities of obtaining a loan. For applications which would normally have been rejected by the bank an activity was added to allow for the applications which initially do not comply with bank requirements to be approved if additional insurance (e.g., another guarantor or another insurance mechanism) is provided. (See Fig. 2.)

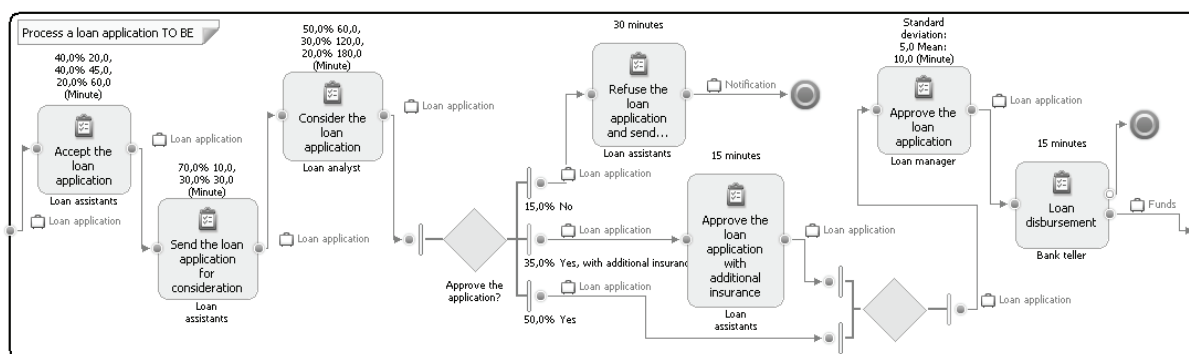


Figure 2. Business process – Processing a loan application TO BE

The analysis showed that the number of such applications, which would be rendered acceptable with additional insurance provided, may amount to around 70% of the applications that were initially rejected. In this way, the bank would approve loans to a significantly larger (approximately 35%) number of clients, in 85% of cases. All improvements are just an example. The improved business process – *Processing a loan application TO BE* is shown in Fig. 2. The results obtained for the improved TO BE model upon performing the 1548 iterations of the simulations, with the same initial parameters of the simulation model, are shown in Table 3.

Table 3. Simulation results in the TO BE state

Process	Duration	Cost	Revenue	Profit
TO BE	3 hrs 6 min	163,04 kn	1.241,19 kn	1.078,15 kn

6. Case study results

By improving the business process model and comparing the results of simulations of AS IS and TO BE states, the following indicators of process modifications in terms of duration, costs, revenues and profit were obtained.

Negative indicators: a) the average process duration time increased by 4% (7 min) – from 2 hours and 59 minutes (179 minutes) to 3 hours and 6 minutes (186 minutes), b) the application processing cost increased by 4% (6,42 kn per application) – from 156,62 kunas to 163, 04 kunas (on average).

Positive indicators: a) on average, the revenue increased by 70% (511,19 kn) – from 730 kunas to 1241,19 kunas (on average), b) the fee is charged for the processing of the approved loans (in the AS IS model this refers to 50%, whereas in the TO BE model this refers to 85% of the cases), c) the profit, that is, the difference between the total costs incurred in the process

and the total revenue, is increased by 88% (an average of 504,77 kn per application) from 573,38 kunas to 1078,15 kunas (on average). By implementing the new process the duration and costs of loan application processing would be only slightly increased. On the other hand, significantly more loans could be approved in this way. Consequently, the bank will charge more loan application processing fees and ensure a considerable profit. To confirm this, we can use an informative calculation of the significance of business process improvement of a bank's overall performance. In the example above, the bank accepts approximately 129 loan applications per month per branch employing one loan assistant. On average, for each application processed in the TO BE model the profit made by the bank exceeds that made in the former AS IS state by approximately 511 kunas. Let us assume that the bank has 100 branches all over Croatia employing 100 loan assistants. It can be concluded that, owing to this improvement, the bank can make very nice increase in profit monthly and annually. Although this is just an example of a process in which an elaborate process model was created taking into consideration just direct costing (labour price), it is evident that business process improvement has

a great impact on the organizational performance. Results are presented in Table 4.

Table 4. Results (profit) in the TO BE process

Profit per application	1 branch office per month	100 branch offices per month	100 branch offices per year
511 kn	65.919 kn	6.591.900 kn	79.102.800 kn

From the process improvement results it is evident that they actually lead to increased profit and better efficiency of a bank's business process, although initially it may appear that they imply an increase in the costs and duration.

7. Simulation and analyzes capabilities of the IBM WebSphere Business Modeler

On the basis of research of a real-world process and the results of the performed simulations, it can be concluded that the described business process modeling tool with simulation capabilities fulfils its function. The following Table 5, contains systematized modeling, simulation and analysis capabilities of the IBM tool.

Table 5. Modeling, simulation and analyses capabilities of IBM WebSphere Business Modeler

	Capability	Description
1. Modeling	BPMN 1.1 compliance	Yes, fully, with extended possibilities of creating a custom artefact (BPMN option)
	Repository support	Yes, all elements can be locally and globally created.
	Structure of models	Hierarchical structure of the model, with possible local sub processes.
	Business items	Content processed in activities, possible full description which enables operationalization of a real data simulation and further deployment of the process in an information system (data entity in a database model).
	Organizational structure	Possible organizational structure, functions, departments, location in which the process runs (fully or partially).
2. Simulation	Activities	Duration can be set stochastically and dynamically (in a function-approximated way), it can also be resource dependent.
	Resources	Three types of resources are available: Roles, Individual (people) and Bulk resources (equipment).
	Gateways	Branching and joining of flows can be made in accordance with the probability of flow or with rules using item attributes values.
	Simulation customizations	In every simulation it is possible to define: start time, time interval between iterations, random (function-approximated) time interval, and interval in or outside the working hours. Simulation virtual running time frame. Number of instances for every input business item (simulation token).
3. Analyses	Static analysis	Covers organizational and resource analysis without simulation. Resources allocation, possible allocation, qualification of certain roles.
	Dynamic analysis	Analyze simulation results by 4 types of analysis: instances, aggregated data, process cases and comparison to other simulation results (AS IS vs. TO BE). Results show a) activities: duration, wait time, cost per time unit, allocated resources, b) resources allocation, usage, usage per activity, cost and duration, c) process general information: average duration, duration, costs, revenue, break events; allocation of resources per process (shows subprocess data separately).

In this way a single improved model – or a few of them – can be compared against the initial model so as to determine the best alternative option for improving a particular business process.

One of the few drawbacks of the tool, where some improvement could still be made, is limited capabilities of dynamically determining the resource operation time spent on activities. Namely, regarding their dynamic duration, a difference between the dynamically determined process duration and the operating time of the resource involved in the process has been noted to occur in some cases.

8. Conclusion

The conducted research and case study example confirms that the IBM WebSphere Business Modeler is applicable to problems related to business process simulation and that its usefulness is beyond question. Tool delivers comprehensible graphical and analytical display of the model simulation results by using standard notation. Initialization and parameterization of a simulation is also fairly simple, but comprehensive.

Regarding the concrete examined capabilities every user, experienced with simulation tools, can get insight into its systematically presented possibilities and conclude its usefulness in their particular case.

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